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Hybrid ML models for volatility prediction in financial risk management

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ABSTRACT

Predicting volatility in financial markets is an important task with practical uses in decision-making, regulation, and academic research. This study focuses on forecasting realized volatility in stock indices using advanced machine learning techniques. We examine three key indices: the Shanghai Stock Exchange Composite (SSE), Infosys (INFY), and the National Stock Exchange Index (NIFTY). To achieve this, we propose a hybrid model that combines optimized Variational Mode Decomposition (VMD) with deep learning methods like Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Using data from 2015 to 2022, we analyse how well these models predict volatility. Our findings reveal distinct patterns: the SSE shows high unpredictability, INFY is prone to extreme positive volatility, and NIFTY is relatively moderate. Among the models tested, the Q-VMD-ANN-LSTM-GRU hybrid model consistently performs best, providing highly accurate predictions for all three indices. This model has practical benefits for financial institutions. It improves risk management, supports investment decisions, and provides real-time insights for traders and risk managers. Additionally, it can enhance stress testing and inspire innovative trading strategies. Overall, our study highlights the potential of advanced machine learning, especially hybrid models, to address financial market complexities and improve risk management practices.

1. Introduction

The landscape of risk management within the realm of finance has undergone a notable transformation through the increasing integration of machine learning (ML)-based risk models (Mashrur et al., 2020). This paradigm shift signifies a profound alteration in the methodologies employed by financial institutions to navigate and mitigate risks. The potential of ML techniques to substantially enhance risk forecasting has garnered substantial attention, with a particular focus on volatility and uncertainty measurement (Ghani et al., 2022; Liang et al., 2020). This enhancement carries significant implications for managing emerging risks inherent to financial assets, thus fortifying the decision-making capabilities of financial institutions, investors, and policymakers (Tien & Hung, 2022; Tripathy, 2022). Volatility, as posited by Markowitz (1967), assumes a pivotal role in risk assessment within financial institutions, wherein the precise estimation of volatility is instrumental for the valuation of financial derivatives and the evaluation of portfolio

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risks. This parameter, originating from the oscillations in asset prices, wields a direct influence over the determination of financial derivatives' prices and the subsequent evaluation of portfolio risk. An accurate projection of volatility assumes paramount significance in deducing derivative values, devising effective hedging strategies, and formulating value-at-risk models, as underscored by Fan et al. (2008). Furthermore, the pertinence of volatility transcends its technical essence, exerting a cascading influence across market regulation and policy-making endeavors, as observed by Chun et al. (2020) and Lu et al. (2021). Noteworthy research contributions by Mei et al. (2017) and Wang et al. (2022) underscore the interconnectedness between stock market volatility and investor behavior, further accentuating its broad-spectrum implications within the financial landscape.

The foundational methodologies in volatility prediction, encompassing the stochastic volatility (SV) model introduced by Taylor (1994) and the generalized autoregressive conditional heteroskedasticity (GARCH) model proposed by Bollerslev (1986), Engle (1982), and Nelson (1991), have offered theoretical clarity and served as pillars in the discipline (Poon & Granger, 2003). However, the reliance of these models on stationary variables has underscored certain limitations, prompting the exploration of more adaptive methodologies (Kim & Won, 2018; Yang et al., 2022). This emergence of intricate and non-linear patterns has necessitated a paradigm shift, wherein the conventional econometric models prove inadequate in effectively modeling and predicting the intricate dynamics of financial data. This inadequacy, as illuminated by López de Prado (2019), is primarily attributed to the linear nature of traditional econometrics, rendering them insufficient for comprehending complex relationships present within modern financial datasets. Consequently, valuable insights embedded within extensive datasets often remain obscured by the limitations of classical approaches, a notion expounded by Coulombe et al. (2021). Contrasting this, ML methodologies possess an inherent adaptability, empowering them to tackle the multifaceted challenges posed by diverse data types and intricate relationships present in contemporary financial data (Rad et al., 2020). The growing utilization of ML techniques for bolstering the precision of volatility projections substantiates the ascendancy of ML methodologies over conventional econometric models (Goodell et al., 2021). This ascendancy has ignited a surge of research interest across diverse financial domains, spanning algorithmic trading (Khurana et al., 2023), asset and derivative pricing (Bilev, 2018), automation (Sokolovsky & Arnaboldi, 2023), financial modeling (Dixon et al., 2020), fraud detection (Bockel-Rickermann et al., 2023), and loan and insurance underwriting (Mabusela-Motsosi et al., 2022).

The successful application of the random forests' algorithm across various disciplines, leading to improvements in forecast accuracy, is a testament to the efficacy of ML approaches (Luong & Dokuchaev, 2018; Vrontos et al., 2021). Furthermore, hybrid models that amalgamate GARCH and ML techniques have exhibited enhanced predictive accuracy for assets such as Bitcoin (Seo & Kim, 2020). Text-mining techniques have also been harnessed to contribute to ML-based volatility forecasting, as demonstrated by Ashtiani and Raahmei (2023). A trend highlighted in recent research involves the deployment of diverse models, including Support Auto Regressive Integrated Moving Average (ARIMA), Vector Machine (SVM), Artificial Neural Network (ANN), XGBoost, and Recurrent Neural Network (RNN), to enhance forecasting accuracy across different applications (Kumar et al., 2021, pp. 877–881). This evolving trend underscores the potential of ML techniques to offer robust solutions in volatility forecasting when compared to traditional econometric models. This trend prompts a reevaluation of the existing strategies within the financial risk management landscape.

This research article presents a comprehensive examination of financial volatility prediction, focusing on the Shanghai Stock Exchange Composite Index (SSE), National Stock Exchange Key Index (NIFTY), and Infosys Limited (INFY) stock, all analysed at 1-min intervals. The study's primary objective is to utilize realized volatility (RV), derived from high-frequency data, and apply an innovative variance decomposition technique to break RV into its constituent Intrinsic Mode Functions (IMFs). These IMFs are then predicted using advanced neural network models, including ANN, Long Short-Term Memory (LSTM) networks, and Gated Recurrent Units (GRU). To further enhance accuracy, a Q-learning algorithm is introduced to amalgamate IMF forecasts, creating a novel ensemble methodology.

This hybrid approach addresses the critical need for precision and sophistication in volatility forecasting, especially within the context of dynamic and fast-evolving financial markets. High-frequency datasets often exhibit complex and nonlinear patterns that traditional econometric models, such as GARCH and SV, struggle to capture due to their reliance on assumptions of linearity and stationarity. These limitations often lead to oversimplified representations of financial time series.

In contrast, the proposed hybrid model integrates Variational Mode Decomposition (VMD) with neural network architectures to overcome these shortcomings. The VMD component isolates latent patterns by decomposing RV into IMFs, mitigating noise and handling nonstationary data with greater precision. The neural network models—ANN, LSTM, and GRU—are well-suited for capturing nonlinear relationships and long-term temporal dependencies, which are critical for understanding high-frequency financial data. The ensemble approach, facilitated by Q-learning, dynamically adjusts the contributions of individual models, leveraging their strengths in specific IMF prediction contexts to enhance overall forecasting accuracy.

This architecture not only improves predictive performance over traditional models but also reveals previously hidden aspects of market dynamics, such as volatility clustering and investor responses to market shocks. These insights offer a more nuanced understanding of the mechanisms driving financial volatility and demonstrate the transformative potential of hybrid machine learning methodologies in financial forecasting. Ultimately, this research introduces a robust framework that strategically combines high-frequency data, advanced variance decomposition, neural network architectures, and ensemble techniques to push the boundaries of volatility forecasting and decision-making in financial markets.

The proposed hybrid model uncovers previously hidden patterns in market behavior that were difficult to discern using traditional econometric methods. By leveraging VMD, the model isolates latent patterns in realized volatility, offering a deeper understanding of the underlying dynamics. Traditional approaches, such as GARCH and SV models, often treat volatility as a homogeneous phenomenon, which limits their ability to disentangle the complex components that contribute to observed market fluctuations. In contrast, VMD decomposes realized volatility into IMFs, each representing distinct frequencies and characteristics. This decomposition enables the identification of nuanced patterns, such as volatility clustering and periodic oscillations, which are often masked by noise in

conventional methodologies.

Moreover, the integration of the Q-learning ensemble technique enhances the model's ability to capture specific investor behaviors and market responses to external stimuli. For instance, the iterative adaptability of Q-learning allows the ensemble to dynamically adjust the weighting of neural network predictions (ANN, LSTM, and GRU) based on their individual strengths in forecasting different IMFs. This flexibility enables the model to detect patterns such as heightened investor sensitivity to sudden volatility shocks or asymmetric responses to positive versus negative news events. These behavioral insights are particularly valuable during periods of market stress, where traditional models may fail to accurately predict abrupt changes in volatility.

The proposed hybrid approach represents a significant advancement in the application of ML techniques to finance, particularly in the realm of volatility forecasting. Unlike traditional econometric models that struggle with nonlinear relationships and temporal dependencies, this methodology integrates VMD with advanced neural network architectures and a Q-learning ensemble framework, creating a robust, dynamic tool for analysing complex financial data. The decomposition of realized volatility into IMFs not only enhances predictive accuracy but also uncovers latent patterns in market behavior, providing deeper insights into volatility dynamics and investor responses. By addressing these gaps, the approach pushes the boundaries of ML's utility in finance, setting a new standard for predictive modelling in high-frequency financial markets.

The broader relevance of this approach extends beyond its immediate predictive capabilities. It demonstrates the transformative potential of hybrid methodologies in advancing ML applications within finance, offering a blueprint for combining domain-specific techniques like VMD with state-of-the-art learning algorithms. This innovative ensemble method can be adapted and scaled to suit various financial contexts, from forecasting volatility in alternative asset classes such as cryptocurrencies and commodities to analysing international financial markets. Its flexibility ensures that it can be customized to capture the unique characteristics of diverse datasets, making it a valuable tool across a wide range of financial applications.

Looking toward long-term goals, the scalability of the hybrid model opens avenues for its integration into automated trading systems, portfolio management platforms, and risk assessment tools. The methodology can also serve as a foundation for cross-market analysis, enabling the simultaneous study of interconnected financial indices to better understand global market dynamics. Furthermore, its adaptability positions it as a cornerstone for future innovations in financial ML, fostering advancements in real-time decision-making, stress testing, and macroprudential policy design. By providing a comprehensive framework that blends precision, adaptability, and scalability, the proposed approach stands out as a groundbreaking contribution to the evolving landscape of machine learning in finance.

The findings also suggest that certain IMFs correlate strongly with specific market events, such as macroeconomic announcements or geopolitical developments. For example, high-frequency IMFs often align with short-term reactions to news events, while low-frequency IMFs reflect underlying market trends and longer-term sentiment shifts.

The structure of this paper is organized as follows: Section 2 provides a brief overview of relevant prior research. Section 3 explains the data sources and outlines the methodology used in this study. In Section 4, we present the empirical findings along with a detailed discussion of the results. Finally, Section 5 concludes the study, highlighting key takeaways and discussing the policy implications of the findings.

2. Review of literature

The integration of Artificial Intelligence (AI) into financial market predictions has brought innovative solutions to forecasting stock prices, volatility, and risk. This review highlights the progress of AI and machine learning (ML) in these areas. Early research mainly focused on predicting market volatility. Engle (1982) introduced the ARCH model, a pioneering method for estimating volatility, followed by Bollerslev (1986), who developed the GARCH model, an improvement that could account for persistent volatility in financial data. Over time, these models were expanded and paired with traditional techniques like linear regression and support vector regression to enhance predictions. The application of neural networks to finance began with Yoon and Swales (1991), who demonstrated their superiority over conventional models in forecasting stock prices for 58 Fortune 500 companies. Haefke and Helmenstein (1996) further validated neural networks, showing that advanced feedforward neural networks improved trading outcomes by better addressing risk and return. Tsaih et al. (1998) extended these ideas by creating a hybrid AI approach that combined neural networks and rule-based systems to predict daily price movements of the S&P 500 index futures. Around the same time, Donaldson and Kamstra (1997) introduced a semi-nonparametric nonlinear GARCH model, inspired by neural networks, which outperformed traditional models like GARCH, EGARCH, and GJR across major markets including New York, London, Tokyo, and Toronto.

In subsequent years, hybrid models continued to evolve. Oh et al. (2006) developed an Early Warning System (EWS) using a mix of ANN and Genetic Algorithms to identify financial crises through daily financial condition indicators. Bildirici and Ersin (2009) built on this work, creating models that integrated ARCH, GARCH, and other machine learning techniques to improve accuracy. Later studies expanded on these hybrid approaches. Bildirici and Ersin (2013), Chen et al. (2013), Fan et al. (2008), Matías et al. (2010), and Wang (2009) explored combining GARCH models with ML techniques, leading to further refinements. Lv et al. (2022) improved the efficiency of ANN using a Search and Rescue optimization algorithm, while Pradeepkumar and Ravi (2017) introduced a Quantile Regression Neural Network trained with the Particle Swarm Optimization method, showing strong results in forecasting financial time series. Recent advancements have also applied hybrid models to emerging assets like cryptocurrencies. For instance, Mtiraoui et al. (2023) used a combination of ARFIMA, empirical wavelet transforms, and local linear wavelet neural networks to predict Bitcoin returns and volatility. Similarly, Tissaoui et al. (2022) demonstrated the predictive power of a Hybrid Particle Swarm Optimization approach for the CBOE volatility index.

Recent trends have also seen the adoption of decomposition methods to improve forecasting accuracy. Shen et al. (2020)

emphasized decomposing realized variance using heterogeneous autoregressive models, while Gao et al. (2023) divided financial asset volatility into upside and downside components, revealing unique predictive characteristics for each. Combining these ideas with deep learning, Li et al. (2021) used variational mode decomposition and a bidirectional long short-term memory neural network to forecast crude oil returns and volatility, highlighting the potential of combining decomposition techniques with advanced machine learning. Overall, the literature reflects significant progress in integrating AI and ML into financial forecasting, particularly through hybrid models that combine traditional and advanced techniques to achieve greater accuracy and reliability.

3. Data and method

3.1. Data

The empirical research conducted here utilizes a comprehensive finance time series dataset spanning from 2015 to 2022. The dataset encompasses three primary variables: Firstly, the minute-to-minute trading data for INFY offers a granular view of the company's stock performance. This variable encapsulates essential metrics like opening and closing prices, highest and lowest prices, as well as trading volume for each minute interval over the specified time frame. Secondly, the minute-by-minute movements of the SSE provide insights into the dynamics of the Chinese stock market. This index serves as a pivotal indicator of the overall performance of the Shanghai stock exchange and its constituent companies. Lastly, the NIFTY index's minute-to-minute changes offer a window into the trends of the Indian equity market. Comprising the performance of the 50 major stocks listed on the National Stock Exchange of India (NSE), NIFTY provides a snapshot of the broader market movements.

The choice of the SSE, NIFTY, and INFY as the focus of this study was guided by their prominence and unique characteristics that make them representative of broader market behavior. These indices were selected to ensure the availability of high-frequency data, which is critical for implementing and evaluating the proposed hybrid methodology. While these indices have been extensively studied in existing literature, they serve as valuable benchmarks for understanding volatility dynamics due to their diverse market structures and investor profiles.

The SSE represents a major emerging market characterized by rapid growth, high volatility, and significant retail investor participation, offering insights into the behavior of developing financial systems. In contrast, NIFTY serves as a proxy for mature emerging markets with a mix of institutional and retail investors, reflecting a relatively stable yet dynamic market environment. INFY, a single stock within the Indian market, provides a micro-level perspective on individual asset behavior, complementing the macro-level analyses of the indices. Together, these three datasets encapsulate a range of market conditions, making them suitable for evaluating the robustness of the hybrid model across different contexts.

3.2. Realized volatility

Based on the method proposed by (Andersen & Bollerslev, 1998), we calculated the intraday RV based on daily returns as follow:

$$RV(t) = \sum_{i=1}^{M} r_{t,i^2} \tag{1}$$

$$r_{t,i} = 100^{*}\left(ln\, P_{t,i} - ln\, P_{t,i-1}\right) \tag{2}$$

3.3. Variational model decomposition (VMD)

In the study by Dragomiretskiy and Zosso (2013), they introduced the VMD algorithm. This methodology is notably more resilient than extant techniques such as the Empirical Mode Decomposition (EMD), especially when confronting sampling inconsistencies and ambient noise. Contrary to EMD, which operates recursively, VMD employs a non-recursive variational mode decomposition paradigm that facilitates the concurrent extraction of modes. This model discerns a conglomerate of modes in conjunction with their pertinent center frequencies. Such identification ensures that the modes, when taken as an ensemble, can accurately replicate the original input signal and maintain their smoothness subsequent to demodulation into the baseband.

The VMD mandates the summation of the bandwidth for each mode, denoted as μ_k. This is expected to be predominantly compact around a central pulsation, ω_k which is identified concomitantly with the decomposition process. This cumulative bandwidth is quantified through the H^1 Gaussian smoothness of the demodulated signal. Specifically, this refers to the squared L^2-norm of the gradient being minimized. This minimization occurs under the stipulated constraint that the aggregate of the modes remains congruent with the initial signal. For a given time series f, the task of decomposition can be represented as addressing the subsequent constrained variational dilemma:

$$\begin{cases} \underset{\{\mu_k\},\{\omega_k\}}{min} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ s.t. \sum_k \mu_k(t) = f(t) \end{cases} \tag{3}$$

where $\{\{\mu\}\}_k := \{\{\mu\}\}_1, \mu_2, \ldots \mu_k\}$ and $\{\{\omega\}\}_k := \{\{\omega\}\}_1, \omega_2, \ldots \omega_k\}$ represents the notations for K nodes and their respective

center frequencies. δ represents the Dirac distribution, t stand for time script, $j^2 = -1$ and * shows the convolution operator. To transform the problem from a constrained to an unconstrained format, a quadratic penalty term is integrated into the equation, accompanied by the introduction of Lagrangian multipliers, represented by λ. The expression for the augmented Lagrangian can then be delineated as:

$$\mathscr{L}(\{u_k\},\{\omega_k\},\lambda): = \alpha\sum_k \| \partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]e^{-j\omega_k t}\|_2^2 + f(t) - \sum_k u_k(t)\|_2^2 + \lambda(t), f(t) - \sum_k u_k(t) \tag{4}$$

In the aforementioned representation, the parameter α is indicative of the balancing factor associated with the data-fidelity constraint. The equation labeled as Eq (4) is resolved utilizing the alternate direction method of multipliers (ADMM). Within this methodology: The mode $uk(\omega)$ in the frequency domain undergoes an update in accordance with Eq. (5). The update equation for center frequencies, represented by ωk, is delineated in Eq. (6). The Lagrangian multiplier, λ, receives an update as per Eq. (7). Additionally, the mode $uk(t)$ in the temporal domain is derived from the real segment of the inverse Fourier transformation of $uk(\omega)$ as specified in Eq. (5)

$$\widehat{u}_k^{n+1}(\omega) = \frac{\widehat{f}(\omega) - \sum_{i\neq k}\widehat{u}_i(\omega) + \frac{\widehat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega-\omega_k)^2} \tag{5}$$

$$\widehat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega|\widehat{u}_k(\omega)|^2 d\omega}{\int_0^\infty |\widehat{u}_k(\omega)|^2 d\omega} \tag{6}$$

$$\widehat{\lambda}^{n+1}(\omega) = \widehat{\lambda}^n(\omega) + \tau\left(\widehat{f}(\omega) - \sum_k \widehat{u}_k^{n+1}(\omega)\right). \tag{7}$$

The minimization function sourced from Scipy's optimize module is employed to reduce the value of the objective function (referred to as 'objective') in relation to the parameter's alpha, tau, and K, while adhering to the specified boundaries. The acronym 'L-BFGS-B' denotes the optimization algorithm Limited-memory Broyden-Fletcher-Goldfarb-Shanno.

3.4. ANN

A simple neural network includes an input later and one or many layers which are hidden and one output layer (Kelleher & Tierney, 2018). There are cells (one or many) in each layer. $y_{i,j}$ equals to the weighted sum of all the inputs to a particular cell. The basic NN uses an activation function which can be described as:

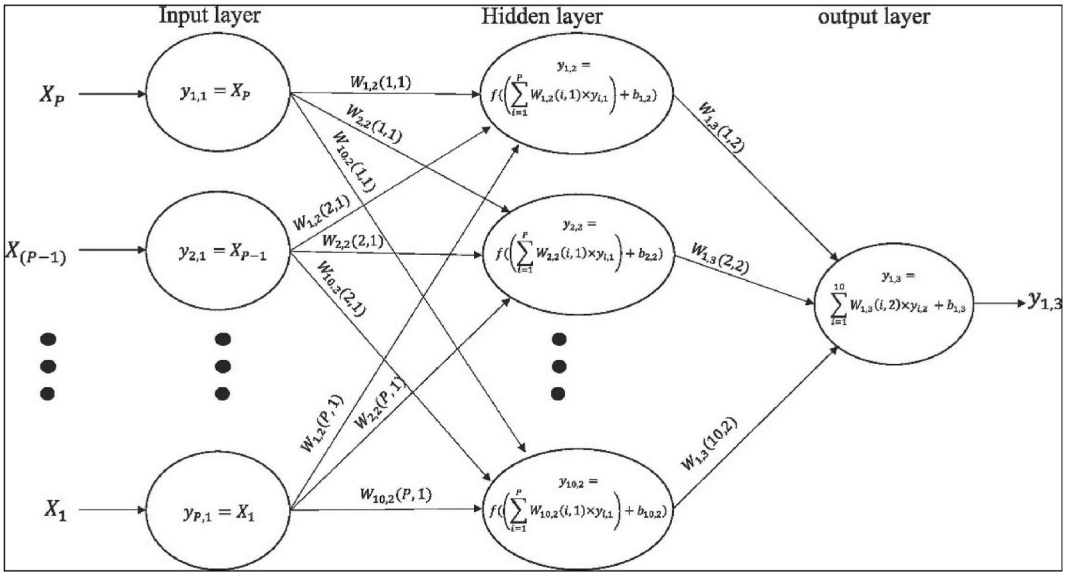


Fig. 1. A diagram depicting a basic Neural Network (NN) configuration is illustrated, comprising of P input cells, a single hidden layer housing 10 sigmoidal cells, and culminating in a sigmoidal output cell. In this context, X_P represents the P_{th} lag of the p. Source: (Solgi et al., 2021).

$$y_{i,j} = f\left(\sum_{l}\left[W_{i,j}(l,j-1)\times y_{l,j-1}\right] + b_{i,j}\right) \tag{8}$$

where $y_{i,j}$ equals to the output of cell i of layer j, $W_{i,j}(l,j-1)$ equals to the weight of the association between cell i of layer j and cell l of layer j-1, and $b_{i,j}$ represents the bias of cell i of layer j. The basic neural network architecture consists of a single hidden layer containing 10 sigmoidal activation units. The input layer is composed of P input units, where P corresponds to the quantity of preceding sequential observations of groundwater (GW) levels. The output layer comprises a singular unit with an identity activation function, as visually represented in Fig. 1. This straightforward neural network is characterized by full connectivity, signifying that each unit within a given layer maintains connections to all units within both the preceding and succeeding layers. The quantification of parameters within the basic neural network is articulated as $(P \times 10) + 21$, where P signifies the count of input units representing the number of past GW level lags. This formulation encapsulates the sum of the products: the number of cells in the hidden layer multiplied by the summation of the number of input cells, the number of cells in the output layer, and an additional unit for offset, further augmented by the number of cells within the output layer. For instance, when the input configuration entails three prior GW level lags, the ensuing simple neural network encompasses a total of 51 parameters, encompassing both connection weights and biases. These parameters are of paramount significance as they necessitate optimization procedures to effectively minimize the predictive error inherent within the network's outcomes.

3.5. LSTM

The LSTM architecture originates from the framework of neural networks. The foundational concept underlying Neural Networks is rooted in emulating the operational principles of biological neurons in humans. Neurons act as recipients of input signals, which are subsequently weighted and aggregated to generate what is referred to as activation input. When this activation input surpasses a certain threshold, an activation function is triggered, yielding an output. The activation function, a mathematical formulation, plays a pivotal role in determining the ultimate output of the neural network by transforming the weighted input into an activated output specific to that input. Visual representation of a neural network structure is provided in Fig. 1.

However, a notable limitation associated with traditional neural networks is their inherent absence of long-term memory. In a fortunate turn, the LSTM model overcomes this constraint, constituting a major advancement in this regard. The distinguishing feature of the LSTM model lies in its incorporation of recurrently interconnected subnetworks, termed memory cell blocks, within its architecture. These memory cell blocks serve as reservoirs for preserving and managing information over extended intervals. Each individual block encompasses one or more self-connected memory cells, accompanied by three pivotal multiplicative components: the input gate, output gate, and forget gate. These constituent elements collaboratively enable the LSTM model to effectively retain and manipulate information over prolonged sequences. The structural layout of a LSTM model is depicted in Fig. 2.

In order to attain a comprehensively structured grasp of the underlying mechanisms within an LSTM model, we will systematically elucidate the operations carried out by its constituent memory cells.

3.5.1. The core idea behind LSTM

The LSTM architecture is founded upon interconnected memory cell blocks, each of which is composed of three sigmoidal layers and an additional layer termed the hyperbolic tangent (tanh) layer. Fig. 3 illustrates the fundamental structure of LSTM memory cells. The initial phase of LSTM operation entails determining which information to exclude, a task undertaken by the sigmoidal layer known as the "forget gate layer." As depicted in Fig. 3, this forgets gate layer evaluates both the output from the preceding input, h_{t-1}, and the external information x_t of the current time step, amalgamating them into an extended vector designated as $[h_{t-1}, x_t]$ through a σ (sigmoid) transformation.

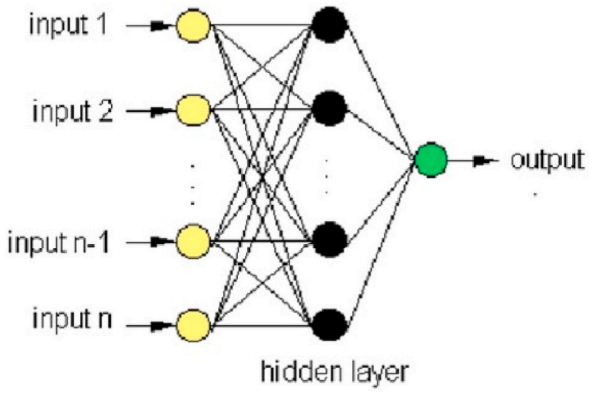


Fig. 2. Structural layout of a LSTM model.

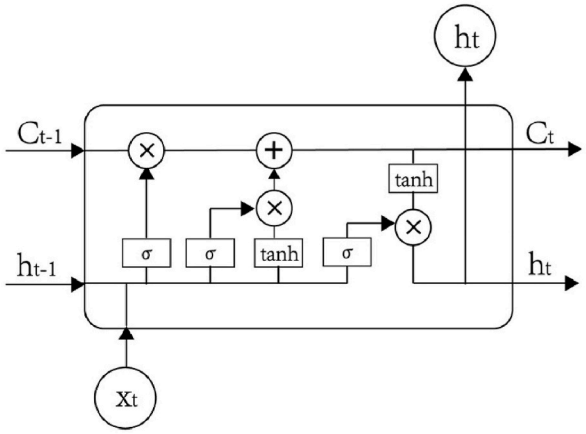


Fig. 3. Architecture of long short-term memory (LSTM). Source: https://doi.org/10.1371/journal.pone.0227222.g001.

$$f_t = \sigma\left(W_f.[h_{t-1}, x_t] + b_f\right), \tag{9}$$

Here, W_f and b_f denote the weight matrix and bias associated with the forget gate, respectively. The sigmoid function $\sigma = \frac{1}{1+e^{-z}}$ is employed, where z represents the input transformation. The core objective of the forget gate lies in quantifying the allocation of the cell state C_{t-1} from the previous time step for integration into the present cell state C_t .The gate produces an output value within the range of 0–1, predicated upon h_{t-1} and x_t. A value of 1 signifies complete retention, whereas a value of 0 signifies total exclusion, as depicted in Fig. 4

The subsequent phase involves determining the specific new information to be incorporated into the cell state. This process unfolds in two distinct components. Initially, a sigmoidal layer referred to as the "input gate layer" is responsible for identifying the facets of the cell state that warrant updating, as delineated in Equation (10). Subsequently, a hyperbolic tangent (tanh) layer generates a vector of novel candidate values, $C_{t,}$ thereby regulating the extent to which fresh information is introduced, as stipulated in Equation (11).

$$i_t = \sigma(W_t.[h_{t-1}, x_t] + b_i) \tag{10}$$

$$\widehat{C_t} = \tanh(W_c.[h_{t-1}, x_t] + b_c) \tag{11}$$

The visual representation of this stage is presented in Fig. 5. Subsequently, the subsequent action entails the revision of the cell state within the memory cells. To achieve this, the prior state C_{t-1} is multiplied by f_t, embodying the determined forgetting process. Following this, the multiplication of $i_t * \widehat{C_t}$ ensues, whereby the newly proposed values are incorporated, their scaling in alignment with the individualized update decisions for each state. This is denoted by Equation (12)

$$C_t = f_t * C_{t-1} + i_t * \widehat{C_t} \tag{12}$$

The concluding phase involves determining the content that will be deemed as output. This is controlled by the output gate, which governs the extent to which the present cell state is disregarded. To initiate this process, a sigmoidal layer is employed to ascertain the output details. Subsequently, the cell state undergoes transformation through the hyperbolic tangent function (tanh), confining the resultant values within the range of −1 to 1. These transformed values are then multiplied by the output derived from the sigmoidal

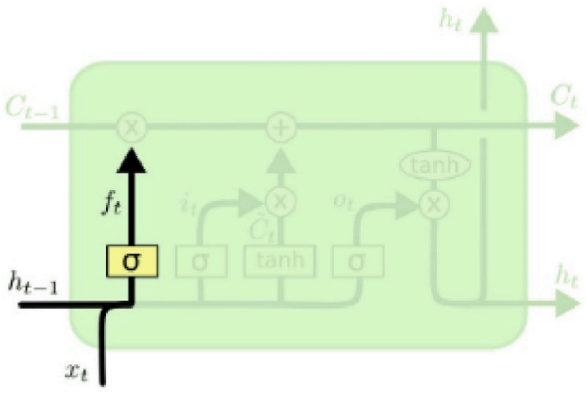


Fig. 4. Visualizing the initial stage of LSTM operation. Source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/.

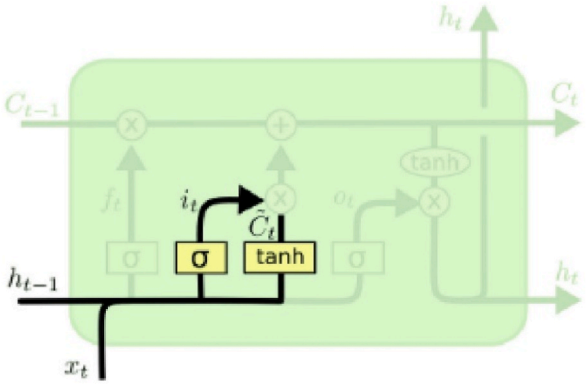


Fig. 5. Visualizing the second stage of LSTM process. Source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/.

layer. This strategic multiplication ensures that only the designated portions are selected for output, as visually depicted in Fig. 6.

$$O_t = \sigma(W_\sigma.[h_{t-1}, x_t] + b_o) \tag{13}$$

The ultimate output value of the cell (Fig. 7) is formulated as follows:

$$h_t = O_t * \tanh(C_t) \tag{14}$$

3.6. GRU

The principal distinction between LSTM and GRU architectures lies in the absence of an independent memory cell within the GRU model. Consequently, the activation h_t^j is derived through the subsequent expression:

$$h_t^j = (1 - z_t^j)h_{t-1}^j + z_t^j \tilde{h}_t^j. \tag{15}$$

The update gate z_t^j determines the degree of update content, influenced by the preceding h_{t-1}^j and the candidate activation $h_t^{\sim j}$. In Equations (16)–(18), the matrices W and U symbolize the weight matrices subject to learning during the training process. Additionally, the subscripts z and r pertain to the update and reset gates respectively. The expressions for obtaining the update gate z_t^j and the reset gate r_t^j are as follows:

$$z_t^j = \sigma(W_z x_t + U_z h_{t-1})^j \tag{16}$$

$$r_t^j = \sigma(W_r x_t + U_r h_{t-1})^j \tag{17}$$

where σ is a sigmoid function. The candidate activation $\tilde{h}_t^j$ is obtained by:

$$\tilde{h}_t^j = tan\, h(W x_t + r_t \odot (U h_{t-1}))^j. \tag{18}$$

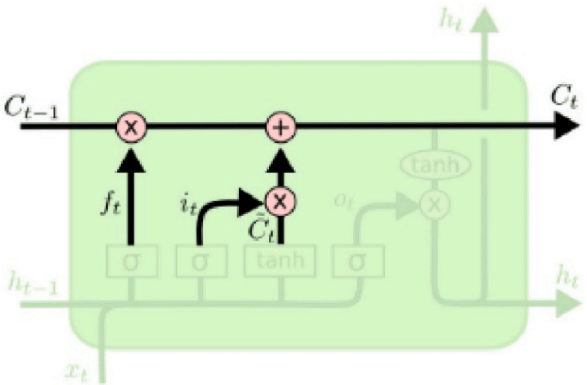


Fig. 6. Visualizing the third stage of LSTM process. Source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/.

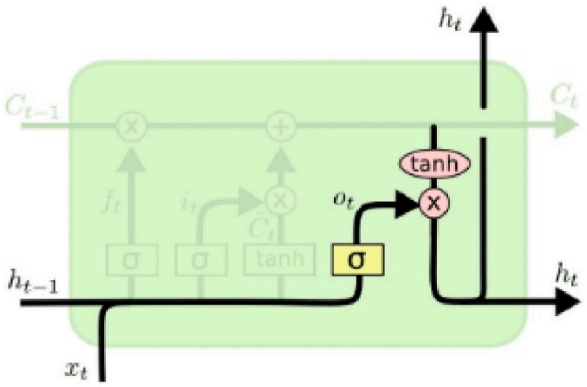


Fig. 7. Visualizing the final stage of LSTM process. Source: https://colah.github.io/posts/2015-08-Understanding-LSTMs/.

where $\odot$ denotes element-wise multiplication.

3.7. Q-learning

The Q-learning technique creates tables, namely the target state matrix 'S' and the action state matrix 'A,' to hold Q-values. These values represent the expected benefits of taking certain actions from specific states. By considering these Q-values, the algorithm chooses the action that offers the highest potential advantage. In our research, we utilize the Q-learning algorithm as a comprehensive learning approach. This involves assigning weights to three deep neural networks. These weights are determined by establishing sensible strategies for rewarding positive outcomes and penalizing negative ones. The procedure for implementing this approach is outlined as follows:

First create a matrix $S = \left[\omega_i^d, \omega_i^l, \omega_i^g\right]$ and matrix $A = \left[\Delta\omega_i^d, \Delta\omega_i^l, \Delta\omega_i^g\right]$. The elements in matrix S represents the weights of the co-efficients from NN, LSTM and GRU. In matrix A, we represent the magnitude of the adjustment action.

Second, build a loss function E and a reward mechanism R:

$$R = \begin{cases} E_t - E_{t+1} - 1, E_t < E_{t+1} \\ E_t - E_{t+1} + 1, E_t \geq E_{t+1} \end{cases} \tag{19}$$

$$E = \sqrt{T^{-1}\sum_{t=1}^{T}\left[\mu_i(t) - y_i(t)\right]^2} \tag{20}$$

where T shows the number of input samples, $y_i(t)$ represents the predicted values of i-th IMF.

In third step, based on ε-greedy mechanism, the agent selects the optimal action with probability of 1- ε, and randomly selects the action with the probability of ε. The action is selected as follows:

$$a_m = \begin{cases} on\ account\ of \max Q, P = 1 - \varepsilon \\ random\ a \in A \end{cases} \tag{21}$$

where $\varepsilon\ \epsilon(0,1)$ which is the exploration probability

In the last step, calculate E and R and update the Q-value table suing the calculated value function $Q_{t+1}(S_t, a_t) = Q_t(S_t, a_t) + \mu\left(r_t + \varnothing max\ Q_t\left(S_{t+1}, a_{t+1}\right) - Q_t(S_t, a_t)\right.$where μ is the learning rate, $\varnothing$ is the discount factor, we need to repeat third and fourth step till the iteration is terminated.

To predict financial market behavior and volatility, this study employs a sophisticated combination of advanced methodologies designed to simplify complex financial data and uncover actionable insights. Financial markets are inherently noisy and unpredictable, with data often obscured by random fluctuations and overlapping patterns. To address these challenges, we utilize VMD, a powerful technique that breaks down intricate financial data into smaller, distinct components known as IMFs. These components allow us to isolate specific patterns or trends within the data, such as short-term market fluctuations or long-term cyclical behaviors, making it easier to identify the underlying drivers of volatility. By separating relevant signals from noise, VMD provides a clearer view of market dynamics and enhances the accuracy of subsequent analyses.

Building on this foundation, we employ ANN, LSTM, and GRU, a family of machine learning models inspired by the way the human brain processes information. ANN, which is widely known for its ability to identify and model relationships in data, provides a solid baseline for predicting future market movements. LSTM, a specialized type of neural network designed for sequential data, excels at recognizing and retaining information over extended time horizons, making it particularly effective for detecting long-term trends and dependencies in financial time series. GRU, another variation of recurrent neural networks, is tailored to handle rapid changes in data efficiently. GRU models are computationally efficient and capable of capturing short-term volatility patterns, making them ideal for analysing fast-evolving market conditions. Together, these architectures enable a comprehensive analysis of both short-term and long-

term dependencies in financial data.

To maximize the strengths of these individual methods, we integrate them into a hybrid model that combines VMD, ANN, LSTM, and GRU into a unified framework. This ensemble approach leverages the complementary capabilities of each component, creating a system that can analyse financial data from multiple perspectives and provide more accurate and reliable predictions. The hybrid model captures complex, nonlinear relationships in market data while simultaneously accounting for noise and temporal dependencies. To further enhance the model's adaptability and performance, we incorporate Q-learning, a reinforcement learning algorithm that dynamically adjusts the weights assigned to different neural networks based on their individual strengths in specific contexts. By learning from new data and adapting to changing market conditions, Q-learning ensures the hybrid model remains effective even in volatile or unpredictable environments.

This combination of methodologies offers several practical advantages. For instance, the hybrid model is adept at identifying key market phenomena, such as volatility clustering, where periods of high market uncertainty are often followed by more uncertainty, and asymmetric market responses, where markets react differently to positive and negative shocks. Additionally, the model can analyse the impact of macroeconomic factors, geopolitical events, and other external influences on market behavior. By translating complex, high-frequency financial data into meaningful patterns and trends, these methods provide valuable insights for decision-makers. Investors and portfolio managers can use the model to anticipate market risks, optimize asset allocations, and develop more effective hedging strategies. Policymakers and regulators can leverage its predictive capabilities to monitor systemic risks, design targeted interventions, and enhance market stability.

In summary, this study demonstrates how advanced machine learning methodologies, particularly the integration of VMD, ANN, LSTM, GRU, and Q-learning, can overcome the limitations of traditional models and provide a deeper understanding of financial markets. By combining cutting-edge techniques with practical applications, this hybrid approach sets a new standard for forecasting market volatility and managing financial risks in an increasingly complex global economy.

4. Empirical results and discussion

Table 1 provided descriptive statistics relate to the realized volatility, which was computed using intraday returns with 1-min intervals of three financial time-series—INFY, SSE, and NIFTY. In the analysis, the SSE manifests the highest average realized volatility at 5.17043. It's followed by INFY, with an average of 1.719876, and then NIFTY, which stands at 0.8804. When it comes to the variability of this realized volatility, the SSE leads in terms of unpredictability, registering the highest standard deviation of 9.5665. Contrastingly, NIFTY shows the least volatility with a standard deviation of 0.749674, whereas INFY finds its volatility intermediate at 2.483301. In the context of extreme values in realized volatility, the SSE has the record for the highest peak at 115.43507. This dwarfs the peak values of both INFY and NIFTY, which reached maximums of 71.368280 and 11.7832 respectively. On the spectrum's opposite end, the lowest realized volatility observed for the SSE was a mere 0.2828, just a smidgen beneath NIFTY's lowest of 0.2989 and substantially less than INFY's 0.587165. As for the shape and distribution of these volatilities, the kurtosis metric points to the "tailedness" of the distribution. Here, INFY's volatility distribution is notably exceptional, with a remarkable kurtosis of 654.597, suggesting very pronounced tails. This is in stark contrast to the SSE 30.36554 and NIFTY's 71.625. Skewness values, which measure distribution asymmetry, unveil that all three indices lean positively. However, INFY stands out starkly, its skewness value of 23.905 hinting at a significant rightward tilt, indicative of more frequent extreme values on the higher side, compared to the SSE 4.59012 and NIFTY's 6.823. In essence, while the realized volatility of the Shanghai Composite Index is marked by its pronounced unpredictability and elevated average values, INFY's stands out due to its significant right-skewed distribution, indicating a propensity for extreme positive volatilities. NIFTY's realized volatility, meanwhile, portrays itself as relatively moderate, especially when juxtaposed with its counterparts.

In the empirical analysis conducted on the realized volatility of three distinct financial time-series - INFY, NIFTY, and SSE - a series of neural network architectures were employed, namely, the ANN, LSTM, and the GRU. For the ANN architecture, it was observed that the design utilized for each of the three financial time-series consisted of two hidden layers. The initial hidden layer comprised 64 units, while the subsequent layer had 32 units. It is noteworthy that the learning rate, pivotal for the pre-training phase of the model, was consistently set at 0.001 across all three indices.

Transitioning to the LSTM model, a recurrent neural network variant tailored for sequence prediction, the consistent optimizer of choice for all indices under study was the Adam optimizer. In an endeavour to mitigate the potential challenges of overfitting, a dropout rate of 0.2 was systematically implemented across the indices. Furthermore, the LSTM structure mirrored that of the ANN in terms of hidden layer configuration, with the first layer containing 64 units and the latter housing 32 units. Lastly, the empirical

Table 1
Descriptive statistics for the RV.

Index	INFYOSYS	SSE	NIFTY
Mean	1.7199	5.1704	0.8804
Std	2.4833	9.5665	0.7497
Maximum	71.3683	115.4351	11.7832
Minimum	0.5872	0.2828	0.2989
Kurtosis	654.5970	30.3655	71.6250
Skewness	23.9050	4.5901	6.8230

research also incorporated the GRU, another recurrent neural network variant. The configurations for this model showcased a uniform approach across all three financial indices. The Adam optimizer was once again the preferred choice, and the dropout rate was consistently maintained at 0.2. The architecture in terms of hidden units remained consistent with both the ANN and LSTM, encapsulating 64 units in the initial layer and 32 in the subsequent layer. In summation, this empirical study underscores a consistent modeling approach across all three financial indices, irrespective of the neural network architecture chosen. Such uniformity in model configuration facilitates a more cogent comparative analysis between the indices (see Table 2).

In the pursuit of predicting the realized volatility of the INFY index, a comprehensive exploration of various machine learning models was conducted, and their efficacy was assessed through the lens of three distinct evaluation metrics: Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The results are presented in Table 3. The conventional ANN model yielded a MAPE of 36.562, accompanied by an MSE of 0.7372 and an RMSE of 0.8586. This performance was closely rivalled by the LSTM model, which demonstrated a slightly improved MAPE of 32.8837, with corresponding MSE and RMSE values of 0.708 and 0.8418. In contrast, the GRU model exhibited a MAPE of 39.0560, intriguingly paired with higher error values represented by an MSE of 0.7859 and an RMSE of 0.8865. The introduction of VMD (see Figs. 8, 9 and 10) into these models introduced nuanced outcomes. The VMD + ANN model showcased a MAPE of 33.418 with MSE and RMSE figures of 0.6995 and 0.8363, while the VMD + LSTM yielded a marginally elevated MAPE of 37.5312 and corresponding MSE and RMSE values of 0.77948 and 0.8828. Similarly, the VMD + GRU model exhibited a MAPE of 35.3814, paired with an MSE of 0.8090 and an RMSE of 0.8995. Notably, the most sophisticated model in this study, the Q-VMD-ANN-LSTM-GRU hybrid, amalgamating Q-learning, VMD, and neural network ensemble, stood out with the lowest MAPE of 30.6542. Its superior performance was further reinforced by the lowest MSE and RMSE values of 0.5087 and 0.7132, respectively. These findings underscore the potential of hybrid models, particularly those integrating Q-learning with VMD and neural network ensembles, in achieving exceptional accuracy in forecasting the realized volatility of financial indices like INFY. In the context of the SSE, the conventional ANN model demonstrated a MAPE of 405.25714, while its corresponding MSE and RMSE were registered at 34.9280 and 5.9099, respectively. Transitioning to the LSTM model, a slightly elevated MAPE of 416.6608 was recorded, alongside an MSE of 39.0946 and RMSE of 6.25257. Conversely, the GRU model manifested a MAPE of 391.7488, accompanied by an almost identical error profile to LSTM, represented by an MSE of 39.04605 and an RMSE of 6.24868. Introducing Variance Mode Decomposition (VMD) led to divergent outcomes. The VMD + ANN model yielded a MAPE of 454.4077, coupled with escalated error values denoted by an MSE of 70.8847 and an RMSE of 8.4193. This trend persisted with the VMD + LSTM, which displayed a MAPE of 486.2014, accompanied by MSE and RMSE values of 77.6498 and 8.81191, respectively. Similarly, the VMD + GRU model presented the highest MAPE of 510.7342, compounded by an MSE of 81.4359 and an RMSE of 9.0242. In contrast, the intricate Q-VMD-ANN-LSTM-GRU model, fusing Q-learning, VMD, and neural network structures, emerged as a standout performer. It showcased the lowest MAPE of 270.456, reflected similarly in its MSE and RMSE values of 30.5645 and 5.5285, respectively. In summary, while traditional and VMD-integrated models offered valuable insights, the hybrid Q-VMD-ANN-LSTM-GRU model demonstrated exceptional proficiency in predicting the realized volatility of the Shanghai Stock Exchange, highlighting the potential of advanced hybrid methodologies in empirical financial forecasting.

Regarding the NIFTY index, the conventional ANN model exhibited a MAPE of 46.92458, with corresponding MSE and RMSE values of 0.400354 and 0.63273. Moving to the LSTM model, a slightly increased MAPE of 48.2089 was observed, accompanied by MSE and RMSE metrics of 0.44571 and 0.66762. Conversely, the GRU model revealed a higher MAPE of 51.6402, accompanied by an MSE of 0.52490 and RMSE of 0.7245. Integrating VMD yielded diverse outcomes across models. Specifically, the VMD + ANN approach resulted in a MAPE of 53.0926, paired with elevated error values in an MSE of 0.5006 and an RMSE of 0.7075. The VMD + LSTM exhibited a slightly reduced MAPE of 48.912236, with MSE and RMSE metrics of 0.424632 and 0.65163. Similarly, the VMD + GRU model displayed the highest MAPE of 54.3598, along with MSE and RMSE values of 0.5237 and 0.7236. In contrast, the hybrid Q-VMD-ANN-LSTM-GRU model, amalgamating Q-learning, VMD, and neural network configurations, showcased exceptional performance with the lowest MAPE of 28.5787. This excellence extended to its MSE and RMSE metrics, registered at 0.2445 and 0.4944, respectively. In conclusion, while conventional and VMD-enhanced models provided foundational insights, the hybrid Q-VMD-ANN-LSTM-GRU model undoubtedly demonstrated unparalleled proficiency in forecasting the realized volatility of the NIFTY index, emphasizing the promise of composite modeling approaches in financial forecasting endeavours.

When juxtaposed with the ANN model, the Q-VMD-ANN-LSTM-GRU showcased an appreciable performance improvement (refer to Table 4). Specifically, MAPE saw an elevation by 16.16%, the MSE was enhanced by 31.00%, and there was a 16.93% betterment in

Table 2
Model and parameters.

Model	Name of parameter	INFY	NIFTY	SSE
ANN	Hidden units	64 for first layer 32 for second layer	64 for first layer 32 for second layer	64 for first layer 32 for second layer
	Rate of Learning during the Pre- Training Phase	0.001	0.001	0.001
LSTM	Name of optimizer	Adam	Adam	Adam
	Rate of dropout	0.2	0.2	0.2
	Hidden units	64 for first layer 32 for second layer	64 for first layer 32 for second layer	64 for first layer 32 for second layer
GRU	Name of the optimizer	Adam	Adam	Adam
	Rate of dropout	0.2	0.2	0.2

Notes: Artificial Neural Network (ANN); Long Short-Term Memory (LSTM); Gated Recurrent Unit (GRU).

Table 3
Empirical results.

	Model	MAPE	MSE	RMSE
INFY	ANN	36.562	0.7372	0.8586
	LSTM	32.8837	0.708	0.8418
	GRU	39.0560	0.7859	0.8865
	VMD + ANN	33.418	0.6995	0.8363
	VMD + LSTM	37.5312	0.77948	0.8828
	VMD + GRU	35.3814	0.8090	0.8995
	Q-VMD-ANN-LSTM-GRU	30.6542	0.5087	0.7132
SSE	ANN	405.2571	34.9280	5.9099
	LSTM	416.6608	39.0946	6.25257
	GRU	391.7488	39.04605	6.24868
	VMD + ANN	454.4077	70.8847	8.4193
	VMD + LSTM	486.2014	77.6498	8.81191
	VMD + GRU	510.7342	81.4359	9.0242
	Q-VMD-ANN-LSTM-GRU	270.456	30.5645	5.5285
NIFTY	ANN	46.92458	0.40035	0.63273
	LSTM	48.2089	0.44571	0.66762
	GRU	51.6402	0.52490	0.7245
	VMD + ANN	53.0926	0.5006	0.7075
	VMD + LSTM	48.912236	0.424632	0.65163
	VMD + GRU	54.3598	0.5237	0.7236
	Q-VMD-ANN-LSTM-GRU	28.5787	0.2445	0.4944

Notes: Artificial Neural Network (ANN); Long Short-Term Memory (LSTM); Gated Recurrent Unit (GRU); Variational Mode Decomposition (VMD); Q Learning-Variational Mode Decomposition-Artificial Neural Network-Long Short-Term Memory-Gated Recurrent Unit (Q-VMD-ANN-LSTM-GRU); Mean Absolute Percentage Error (MAPE); Mean Squared Error (MSE); Root Mean Squared Error (RMSE).

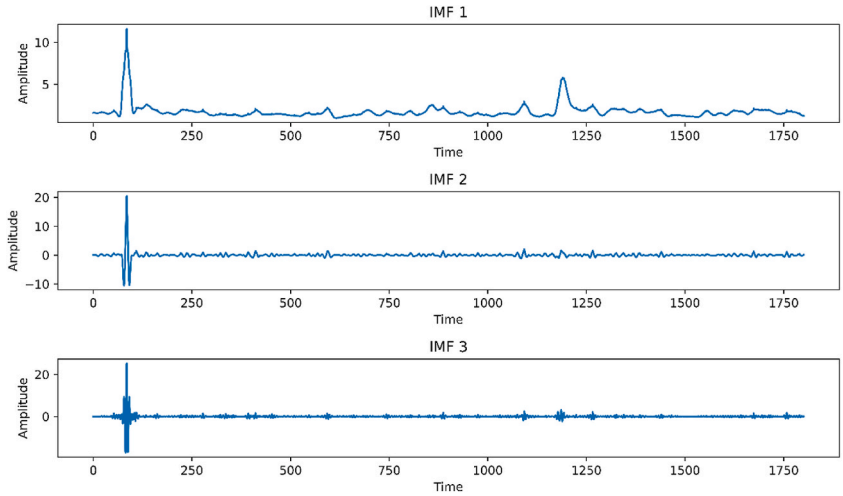


Fig. 8. VMD decomposition of INFY.

RMSE. In comparison with the LSTM model, the Q-VMD-ANN-LSTM-GRU yielded a 6.78% improvement in MAPE. This positive trend was reflected in the MSE, which showed a 28.15% increment, and the RMSE that improved by 15.28%. When set against the GRU model, the Q-VMD-ANN-LSTM-GRU exhibited the most significant MAPE enhancement, registering at 21.51%. Concurrently, the MSE witnessed an upswing of 35.27%, and the RMSE surged by 19.55%. Against the VMD + ANN model, the Q-VMD-ANN-LSTM-GRU model recorded an 8.27% rise in MAPE. The MSE augmented by 27.28%, and the RMSE grew by 14.72%. Benchmarked against the VMD + LSTM model, the Q-VMD-ANN-LSTM-GRU displayed an 18.32% betterment in MAPE. This improvement trajectory continued with the MSE, which elevated by 34.74%, and the RMSE, which escalated by 19.21%. Lastly, when the Q-VMD-ANN-LSTM-GRU model was contrasted with VMD + GRU, it demonstrated a 13.36% improvement in MAPE. This was complemented by a significant enhancement in MSE by 37.12% and an upward movement in RMSE by 20.71%. In summation, the above analysis underscores the augmented precision of the Q-VMD-ANN-LSTM-GRU model in predicting INFY's realized volatility, as it consistently outperformed the other models across all evaluative metrics.

In the empirical analysis pertaining to the SSE realized volatility prediction, when set against the ANN model, the Q-VMD-ANN-LSTM-GRU revealed its superior efficacy, with MAPE witnessing an enhancement of 33.26%. This was accompanied by a 12.49% improvement in MSE and a 6.45% increment in RMSE. Contrasted with the LSTM model, the Q-VMD-ANN-LSTM-GRU illustrated a

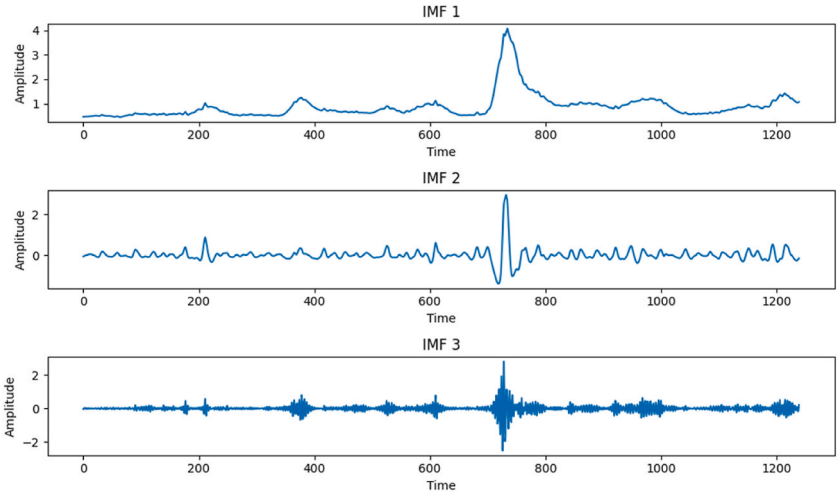


Fig. 9. VMD decomposition of NIFTY.

noteworthy performance elevation. Specifically, MAPE rose by 35.09%, the MSE was enhanced by 21.82%, and the RMSE surged by 11.58%. Benchmarked against the GRU model, the Q-VMD-ANN-LSTM-GRU exhibited an elevation of 30.96% in MAPE. Further, the MSE and RMSE witnessed respective increments of 21.72% and 11.53%. In comparison to the VMD + ANN model, the Q-VMD-ANN-LSTM-GRU model's superiority became even more apparent. The MAPE saw a significant boost of 40.48%. This was complemented by a remarkable increase in MSE by 56.88% and a 34.34% improvement in RMSE. When juxtaposed with the VMD + LSTM model, the Q-VMD-ANN-LSTM-GRU model showcased a considerable improvement in MAPE, registering at 44.37%. Additionally, the MSE grew by 60.64%, and RMSE expanded by 37.26%. Lastly, against the VMD + GRU model, the Q-VMD-ANN-LSTM-GRU illustrated its highest comparative gains. The MAPE soared by 47.05%, accompanied by a 62.47% upswing in MSE and a 38.74% escalation in RMSE. In conclusion, the presented empirical results underscore the Q-VMD-ANN-LSTM-GRU model's commendable precision in predicting the realized volatility of the SSE. Across the board, this model consistently demonstrated enhanced performance over other tested models in all the employed evaluation metrics.

For the NIFTY index, When the Q-VMD-ANN-LSTM-GRU model was pitted against the ANN model, it showcased remarkable superiority. The model recorded a substantial improvement in MAPE by 39.10%. This was closely followed by a 38.93% enhancement in MSE and a notable 21.86% rise in RMSE. Upon contrasting with the LSTM model, the Q-VMD-ANN-LSTM-GRU model further asserted its dominance. Specifically, the improvement in MAPE was registered at 40.72%, while the MSE and RMSE surged by 45.14% and 25.95% respectively. When compared with the GRU model, the Q-VMD-ANN-LSTM-GRU model's proficiency was even more evident, with MAPE recording an increment of 44.66%. In tandem, MSE and RMSE observed upswings of 53.42% and 31.76% respectively.

In the face-off against the VMD + ANN model, the Q-VMD-ANN-LSTM-GRU model continued its trend of outperformance. Here, MAPE witnessed a boost of 46.17%, complemented by rises in MSE and RMSE by 51.16% and 30.12%, respectively. When benchmarked against the VMD + LSTM model, the Q-VMD-ANN-LSTM-GRU model highlighted a MAPE enhancement of 41.57%. Additionally, improvements in MSE and RMSE were noted at 42.42% and 24.13%, respectively. Lastly, in the comparison with the VMD + GRU model, the Q-VMD-ANN-LSTM-GRU model exhibited its most pronounced gains. The MAPE soared by a remarkable 47.43%. Concurrently, the MSE and RMSE escalated by 53.31% and 31.67% respectively. In summation, the empirical findings presented make a compelling case for the Q-VMD-ANN-LSTM-GRU model's efficacy in predicting the NIFTY index's realized volatility. Across all metrics and benchmarks, this model consistently outperformed its peers, further cementing its status as a reliable tool for such predictions.

In the context of robustness analysis, as depicted in Table 5, the forecasting performance of various models is examined across different horizons (3 and 21) for three distinct financial indices: INFY, Shanghai Stock Exchange, and NIFTY. The evaluation is gauged through two pivotal metrics, namely the MAPE and the RMSE, which offer insights into the accuracy and reliability of the models' predictions. For the INFY index, it is observed that across both horizons, the ANN model yields a MAPE of 33.5787 and 38.6079, accompanied by corresponding RMSE values of 0.7773 and 0.82411, respectively. The LSTM model demonstrates slightly higher MAPE values of 37.3691 and 37.3180, alongside comparable RMSE figures of 0.7892 and 0.7614. Notably, the GRU model exhibits the lowest MAPE for both horizons, recording values of 30.4306 and 37.0176, with associated RMSE metrics of 0.7737 and 0.76311. Upon the introduction of VMD, the models' performance is generally affected, leading to higher MAPE and RMSE values. Transitioning to the SSE, the ANN model demonstrates a MAPE of 467.7436 and 430.25803 for the respective horizons, paired with RMSE values of 5.7414 and 7.5648. Similarly, the LSTM model yields MAPE values of 449.0902 and 453.299, accompanied by RMSE metrics of 5.62937 and 7.2818. The GRU model follows suit with MAPE values of 429.84924 and 459.1080, alongside RMSE values of 5.55345 and 7.5878. Interestingly, VMD incorporation results in higher MAPE and RMSE values across the board, implying a potential trade-off between model complexity and prediction accuracy. For the NIFTY index, the ANN model presents MAPE values of 47.07929 and 39.5130, accompanied by RMSE metrics of 0.6499 and 0.43897 for horizons 3 and 21, respectively. The LSTM model exhibits higher MAPE

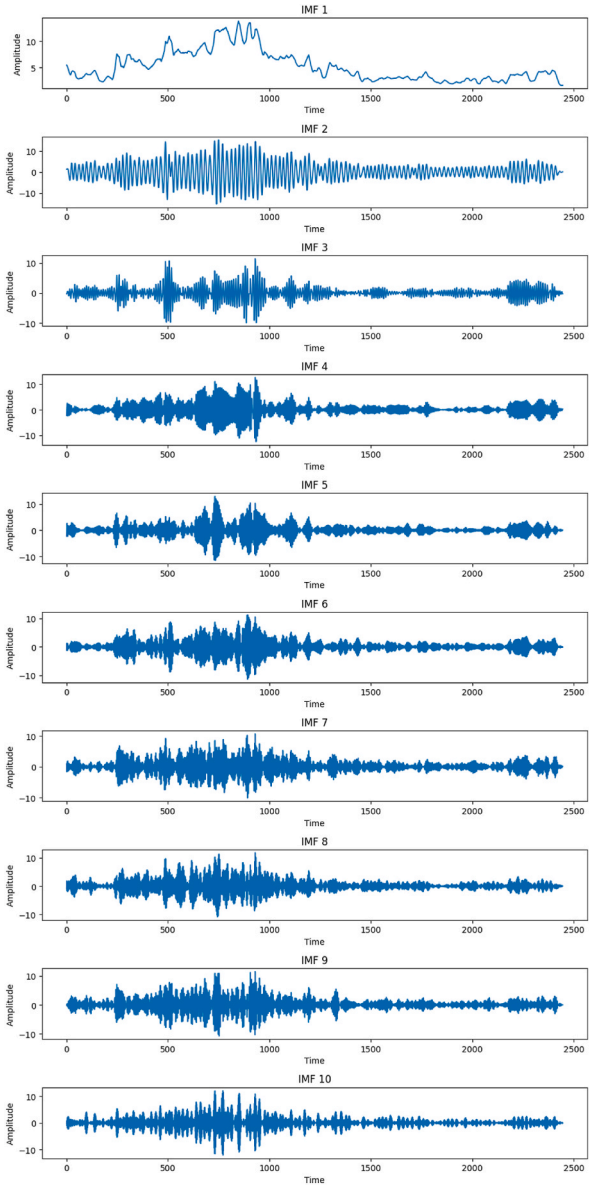


Fig. 10. VMD decomposition of SSE.

values of 55.0420 and 42.49345, coupled with RMSE values of 0.70978 and 0.4341. Remarkably, the GRU model demonstrates identical MAPE and RMSE values of 56.30982 for both horizons. With VMD incorporation, the models generally show elevated MAPE and RMSE values.

Intriguingly, the Q-VMD-ANN-LSTM-GRU hybrid model emerges as a noteworthy contender across indices and horizons, consistently displaying lower MAPE and RMSE values compared to other models. These findings collectively underscore the intricate interplay between model types, dataset characteristics, and prediction horizons in empirical financial forecasting scenarios.

5. Conclusion and policy implications

In conclusion, this empirical research has provided a comprehensive exploration into the intersection of AI, machine learning, and financial market volatility prediction. By delving into various neural network architectures and hybrid models, our study has illuminated the predictive capabilities of these methodologies in forecasting the realized volatility of financial indices, with a specific focus on INFY, SSE, and NIFTY. The empirical insights drawn from our extensive dataset spanning 2015 to 2022 offer valuable perspectives on index behavior and the potential of advanced forecasting techniques. Our analysis of descriptive statistics revealed distinct volatility patterns across the three indices: the SSE exhibited pronounced unpredictability and elevated average values, while INFY

Table 4
Percentage improvement in models.

		Percentage improvement in MAPE	Percentage improvement in MSE	Percentage improvement in RMSE
INFY	Q-VMD-ANN-LSTM-GRU versus ANN	16.16%	31.00%	16.93%
	Q-VMD-ANN-LSTM-GRU versus LSTM	6.78%	28.15%	15.28%
	Q-VMD-ANN-LSTM-GRU versus GRU	21.51%	35.27%	19.55%
	Q-VMD-ANN-LSTM-GRU versus VMD + ANN	8.27%	27.28%	14.72%
	Q-VMD-ANN-LSTM-GRU versus VMD + LSTM	18.32%	34.74%	19.21%
	Q-VMD-ANN-LSTM-GRU versus VMD + GRU	13.36%	37.12%	20.71%
SSE	Q-VMD-ANN-LSTM-GRU versus ANN	33.26%	12.49%	6.45%
	Q-VMD-ANN-LSTM-GRU versus LSTM	35.09%	21.82%	11.58%
	Q-VMD-ANN-LSTM-GRU versus GRU	30.96%	21.72%	11.53%
	Q-VMD-ANN-LSTM-GRU versus VMD + ANN	40.48%	56.88%	34.34%
	Q-VMD-ANN-LSTM-GRU versus VMD + LSTM	44.37%	60.64%	37.26%
	Q-VMD-ANN-LSTM-GRU versus VMD + GRU	47.05%	62.47%	38.74%
NIFTY	Q-VMD-ANN-LSTM-GRU versus ANN	39.10%	38.93%	21.86%
	Q-VMD-ANN-LSTM-GRU versus LSTM	40.72%	45.14%	25.95%
	Q-VMD-ANN-LSTM-GRU versus GRU	44.66%	53.42%	31.76%
	Q-VMD-ANN-LSTM-GRU versus VMD + ANN	46.17%	51.16%	30.12%
	Q-VMD-ANN-LSTM-GRU versus VMD + LSTM	41.57%	42.42%	24.13%
	Q-VMD-ANN-LSTM-GRU versus VMD + GRU	47.43%	53.31%	31.67%

Notes: Artificial Neural Network (ANN); Long Short-Term Memory (LSTM); Gated Recurrent Unit (GRU); Variational Mode Decomposition (VMD); Q Learning-Variational Mode Decomposition-Artificial Neural Network-Long Short-Term Memory-Gated Recurrent Unit (Q-VMD-ANN-LSTM-GRU); Mean Absolute Percentage Error (MAPE); Mean Squared Error (MSE); Root Mean Squared Error (RMSE).

Table 5
Robustness analysis.

		Horizon = 3		Horizon = 21	
		MAPE	RMSE	MAPE	RMSE
INFY	ANN	33.5787	0.7773	38.6079	0.82411
	LSTM	37.3691	0.7892	37.3180	0.7614
	GRU	30.4306	0.7737	37.0176	0.76311
	VMD + ANN	44.4214	1.3163	41.027	0.8044
	VMD + LSTM	43.9539	1.5567	35.8008	0.73123
	VMD + GRU	44.7821	1.9193	33.9204	0.7227
	Q-VMD-ANN-LSTM-GRU	29.4534	0.5687	30.5648	0.5895
SSE	ANN	467.7436	5.7414	430.25803	7.5648
	LSTM	449.0902	5.62937	453.299	7.2818
	GRU	429.84924	5.55345	459.1080	7.5878
	VMD + ANN	388.7430	7.4882	447.959	7.7412
	VMD + LSTM	472.9978	8.00927	446.6018	6.7184
	VMD + GRU	444.5190	8.1640	407.6690	5.9968
	Q-VMD-ANN-LSTM-GRU	310.457	4.587	331.5671	4.4518
NIFTY	ANN	47.07929	0.6499	39.5130	0.43897
	LSTM	55.0420	0.70978	42.49345	0.4341
	GRU	56.30982	0.6999	56.30982	0.6999
	VMD + ANN	57.581	0.7947	48.3861	0.6201
	VMD + LSTM	61.7134	0.92496	36.8349	0.4984
	VMD + GRU	63.222	0.9563	45.8283	0.6017
	Q-VMD-ANN-LSTM-GRU	40.5190	0.5134	31.5689	0.4167

Notes: Artificial Neural Network (ANN); Long Short-Term Memory (LSTM); Gated Recurrent Unit (GRU); Variational Mode Decomposition (VMD); Q Learning-Variational Mode Decomposition-Artificial Neural Network-Long Short-Term Memory-Gated Recurrent Unit (Q-VMD-ANN-LSTM-GRU); Mean Absolute Percentage Error (MAPE); Mean Squared Error (MSE); Root Mean Squared Error (RMSE).

displayed a significant right-skewed distribution indicative of a propensity for extreme positive volatilities. In contrast, NIFTY's volatility emerged as relatively moderate, presenting a diverse volatility landscape.

Our investigation into neural network architectures, including ANN, LSTM, and GRU, underscored the rich landscape of predictive

methodologies. Each model, with its distinct configurations and optimization methods, demonstrated varying performance across the indices. Importantly, the Q-VMD-ANN-LSTM-GRU hybrid model consistently emerged as an exceptional performer, showcasing superior accuracy and precision in volatility prediction. The integration of VMD into these models further deepened our understanding of their robustness and limitations. While VMD introduced nuanced outcomes, the Q-VMD-ANN-LSTM-GRU hybrid model consistently demonstrated remarkable proficiency, surpassing other models in all evaluation metrics. The model's precision in predicting the realized volatility of INFY, SSE, and NIFTY indices underscores its potential applicability in diverse financial contexts.

The findings of this research highlight several practical implications for real-world financial applications, particularly in portfolio management, risk mitigation, and policymaking. The hybrid Q-VMD-ANN-LSTM-GRU model's superior accuracy in predicting realized volatility offers tangible benefits for optimizing portfolio strategies. By providing precise forecasts, the model enables portfolio managers to dynamically adjust asset allocations to achieve better risk-adjusted returns. For example, its ability to detect volatility clustering and short-term fluctuations supports proactive rebalancing of portfolios during high-risk periods, thereby safeguarding returns. Additionally, the granular insights obtained from the decomposition of volatility into IMFs allow for the development of more targeted hedging strategies. Portfolio managers can construct precise combinations of derivatives, such as options and futures, to mitigate exposure to anticipated risks effectively. These predictive capabilities also support algorithmic trading systems by enabling them to identify short-term inefficiencies and adjust execution strategies, reducing market slippage and optimizing trading performance.

In the realm of risk mitigation, financial institutions can leverage the model to enhance their risk management frameworks. The model's accurate volatility decomposition provides a robust foundation for stress testing by simulating the effects of external shocks, such as geopolitical events or macroeconomic announcements, on market volatility. This facilitates the evaluation of portfolio resilience under adverse scenarios. Furthermore, the model's minute-level predictions enable real-time risk monitoring, which is particularly valuable for high-frequency trading desks and risk management teams that require immediate responses to evolving market conditions. Insights from the model can also guide the optimization of capital allocation strategies, allowing financial institutions to adjust liquidity reserves or leverage ratios during periods of heightened volatility to maintain operational stability and compliance with regulatory requirements.

For policymakers, the hybrid model offers a powerful tool to anticipate and manage systemic risks, ensuring market stability. The ability to forecast volatility clusters provides early warnings of potential market disruptions, enabling regulators to implement preemptive measures such as circuit breakers or liquidity injections to curb excessive market turbulence. Moreover, the model uncovers nuanced investor responses to volatility shocks, offering insights that can inform the design of regulatory measures aimed at promoting market transparency and curbing herd behavior during periods of stress. By identifying the long-term effects of macroeconomic policies through low-frequency volatility components, the model aids in formulating macroprudential policies that address systemic vulnerabilities. These applications demonstrate how the findings of this study can be translated into actionable strategies to enhance decision-making, mitigate financial risks, and ensure the stability of dynamic financial markets.

Furthermore, the practical implications extend to real-time decision-making in financial markets. Equipping traders and risk managers with timely insights, the Q-VMD-ANN-LSTM-GRU model empowers informed decisions on position sizing, order execution, and risk exposure adjustments. This proactive approach leverages machine learning to capitalize on market opportunities while managing potential downsides. The model's performance also provides a foundation for robust stress testing and scenario analysis frameworks, enabling financial institutions to fortify their portfolios against market shocks and regulatory requirements. Additionally, the Q-VMD-ANN-LSTM-GRU model's consistency in volatility forecasting makes it an invaluable tool for model-driven risk assessment processes, risk measurement, pricing, capital allocation, and hedging strategies. The potential extends to innovative trading strategies in the hedge fund sector, where the model's precision in volatility forecasting can be harnessed to develop algorithmic trading systems that exploit volatility patterns and market inefficiencies.

In summary, the empirical insights garnered from the Q-VMD-ANN-LSTM-GRU hybrid model reverberate through the realms of machine learning and risk management in finance. The model's precise volatility predictions open avenues for portfolio diversification, tailored hedging mechanisms, optimized risk-return profiles, and informed real-time decisions. These practical implications underscore the substantial promise of advanced machine learning methodologies, particularly when integrated into hybrid models, in navigating the complexities of financial markets and advancing risk management practices.

Author statement

No conflict of interest to declare.

Declaration of competing interest

The authors state that they do not have any known competing financial interests or personal relationships that could have potentially influenced the work presented in this paper.

Appendix A1

Model	MAPE	MSE	RMSE
SPDR S&P 500 ETF Trust			
ANN	100.4998	18.0788	4.2519
LSTM	104.4080	19.0761	4.3676
GRU	96.7075	19.5075	4.4167
VMD + ANN	107.1592	22.1666	4.7081
VMD + LSTM	108.7759	22.5742	4.7512
VMD + GRU	105.1048	22.3459	4.7271
Q-VMD-ANN-LSTM-GRU	29.402	0.4157	0.6534

Data availability

Data will be made available on request.

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